

Experiment 5: Selection Sort – Small Dataset in a Mobile App

Aim

To implement the **Selection Sort** algorithm in Python to sort a small list of product prices and measure the execution time.

Problem Statement

A shopping app needs to sort a **recently viewed list of 8–10 product prices** whenever the user opens the screen. Since the list is very small, implement **Selection Sort** and measure the execution time to show that it is suitable for small datasets.

Algorithm

Step 1: Start.

Step 2: Read the number of product prices.

Step 3: Read the product prices.

Step 4: Record the start time.

Step 5: Apply Selection Sort to sort the prices in ascending order.

Step 6: Record the end time.

Step 7: Display the sorted prices.

Step 8: Display the execution time.

Step 9: Stop.

Source Code (Python)

main.py

```
1  # Selection Sort with Execution Time
2
3  import time
4
5  n = int(input("Enter number of products: "))
6
7  prices = []
8
9  print("Enter product prices:")
10 for i in range(n):
11     prices.append(int(input()))
12
13 start = time.time()
14
15 for i in range(n - 1):
16     min_index = i
17
18     for j in range(i + 1, n):
19         if prices[j] < prices[min_index]:
20             min_index = j
21
22     prices[i], prices[min_index] = prices[min_index], prices[i]
23
24 end = time.time()
25
26 print("Sorted Prices:", prices)
27 print("Execution Time:", end - start, "seconds")
```

Output

Output

```
Enter number of products: 8
Enter product prices:
499
129
899
45
275
60
310
150
Sorted Prices: [45, 60, 129, 150, 275, 310, 499, 899]
Execution Time: 2.2172927856445312e-05 seconds

=== Code Execution Successful ===
```

Complexity Analysis

Aspect	Complexity
Time Complexity	$O(n^2)$ (for $n \leq 10$)
Space Complexity	$O(1)$

Result

The **Selection Sort** algorithm was successfully implemented in Python to sort a small list of product prices. The program displayed the sorted list and the execution time. Since the dataset is very small, the execution time is negligible, making Selection Sort a suitable choice for this application.